

Restorasi Dokumen Berorientasi HTR Menggunakan GAN dengan Diskriminator Dual-Modal dan Optimasi Fungsi Loss

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Ringkasan—Dokumen tulisan tangan historis mengalami berbagai artefak degradasi termasuk noise latar belakang, bleeding tinta, penuaan kertas, dan kerusakan fisik yang sangat berdampak pada sistem Handwritten Text Recognition (HTR). Meskipun metode peningkatan kualitas dokumen yang ada berfokus pada perbaikan kualitas visual, metode tersebut sering gagal mempertahankan keterbacaan teks untuk mesin HTR. Paper ini mengusulkan kerangka restorasi dokumen berorientasi HTR berbasis Generative Adversarial Networks (GAN) dengan diskriminator dual-modal dan optimasi fungsi loss multi-komponen. Berbeda dengan pendekatan konvensional yang hanya menggunakan supervisi tingkat gambar, metode kami mengintegrasikan recognizer HTR terlatih yang di-frozen dalam arsitektur GAN untuk mengoptimalkan keterbacaan teks secara eksplisit selama proses restorasi. Diskriminator dual-modal menggunakan jalur CNN dan LSTM untuk menilai kualitas visual dan koherensi teks sekuensial secara bersamaan. Kami memperkenalkan fungsi loss berorientasi HTR yang menggabungkan adversarial loss, pixel-level reconstruction loss (L1), perceptual loss, dan recognition feature loss yang diturunkan dari recognizer yang di-frozen. Eksperimen ekstensif pada dataset terdegradasi sintesis dan dokumen historis real dari Arsip Nasional Republik Indonesia (naskah paleografi abad 16-18) menunjukkan bahwa pendekatan kami mencapai PSNR sebesar [XX.XX] dB, SSIM sebesar [0.XX], dan mengurangi Character Error Rate (CER) sebesar [XX]% dibandingkan dengan metode state-of-the-art. Hasil menunjukkan bahwa optimasi HTR eksplisit selama training menghasilkan preservasi teks yang superior sambil mempertahankan kualitas visual, menjadikan dokumen hasil restorasi lebih sesuai untuk alur kerja transkripsi otomatis.

Index Terms—Generative Adversarial Networks, Restorasi Dokumen, Pengenalan Tulisan Tangan, Diskriminator Dual-Modal, Optimasi Fungsi Loss, Pemrosesan Dokumen Historis, Peningkatan Kualitas Gambar, Deep Learning.

I. PENDAHULUAN

DOKUMEN tulisan tangan historis merepresentasikan warisan budaya yang tak ternilai dan mengandung informasi kritis tentang peristiwa masa lalu, pemerintahan, perdagangan, dan struktur sosial. Arsip Nasional Republik Indonesia (ANRI) menyimpan koleksi ekstensif dokumen era kolonial dari abad ke-16 hingga ke-18, banyak di antaranya diakui oleh UNESCO sebagai bagian dari Memory of the World Register. Dokumen-dokumen ini, ditulis dalam skrip paleografi Belanda selama

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[PLACEHOLDER GAMBAR 1]

Contoh Degradasi

- (a) Noise latar, (b) Ink bleeding
(c) Penuaan kertas, (d) Kerusakan fisik
(e) Artefak scanning, (f) Kombinasi

Gambar 1: Contoh jenis degradasi yang umum ditemukan pada dokumen tulisan tangan historis dari koleksi ANRI (abad ke-16 hingga ke-18).

periode VOC (Vereenigde Oostindische Compagnie) dan Hindia Belanda, memberikan wawasan unik tentang perkembangan sejarah Indonesia.

Namun, dokumen-dokumen ini mengalami degradasi parah akibat berbagai faktor: penuaan kertas (penguningan, kerapuhan), kondisi lingkungan (kelembaban, fluktuasi suhu), kerusakan fisik (sobekan, kerutan, noda), deteriorasi tinta (bleeding, pudar), dan kondisi penyimpanan yang tidak optimal. Selain itu, proses digitalisasi menimbulkan artefak seperti bayangan, blur, pencahayaan tidak merata, dan noise scanning. Gambar 1 mengilustrasikan berbagai jenis degradasi yang umum diamati pada dokumen historis.

Artefak degradasi ini sangat mengganggu performa sistem Handwritten Text Recognition (HTR), yang esensial untuk digitalisasi skala besar, transkripsi, pencarian, pengindeksan, dan ekstraksi informasi dari arsip historis. Meskipun kemajuan signifikan telah dicapai dalam teknologi HTR menggunakan pendekatan deep learning, khususnya dengan Convolutional Recurrent Neural Networks (CRNN) dan arsitektur berbasis Transformer, sistem-sistem ini tetap sangat sensitif terhadap kualitas gambar dan tingkat degradasi.

A. Keterbatasan Pendekatan yang Ada

Metode binarisasi dan peningkatan kualitas dokumen tradisional dapat dikategorikan secara luas menjadi tiga generasi:

Metode Klasik Pemrosesan Gambar: Pendekatan awal menggunakan teknik thresholding (global atau adaptif), operasi morfologi, dan kerangka kerja minimisasi energi. Metode representatif termasuk thresholding Otsu [1], thresholding adaptif lokal Sauvola [2], dan pendekatan yang lebih canggih seperti yang dievaluasi dalam kompetisi DIBCO (Document Image Binarization Contest). Meskipun efisien secara komputasi, metode-metode ini sangat sensitif terhadap tingkat

degradasi dan penyetelan parameter, sering gagal pada dokumen yang sangat terdegradasi dengan latar belakang yang kompleks.

Metode Berbasis Machine Learning: Pendekatan supervised learning, khususnya menggunakan Support Vector Machines (SVM) dan Random Forests, mencoba mengklasifikasikan piksel sebagai foreground atau background berdasarkan fitur yang dibuat secara manual. Metode-metode ini menunjukkan ketahanan yang lebih baik tetapi masih kesulitan dengan degradasi parah dan memerlukan feature engineering yang ekstensif.

Metode Berbasis Deep Learning: Pendekatan terkini memanfaatkan deep neural networks, khususnya:

- *Arsitektur auto-encoder* (varian U-Net) untuk translasi gambar-ke-gambar end-to-end [5], [6]
- *Generative Adversarial Networks (GAN)* untuk peningkatan kualitas dokumen yang realistis [7], [9]
- *Jaringan cascaded dan multi-stage* untuk penyempurnaan progresif [8]

Meskipun mencapai peningkatan kualitas visual yang mengesankan, metode-metode ini memiliki keterbatasan kritis:

- 1) **Kurangnya optimasi berorientasi HTR:** Metode yang ada mengoptimalkan metrik kesamaan visual (PSNR, SSIM, F-measure) antara gambar hasil restorasi dan ground truth, hanya menggunakan supervisi tingkat piksel. Mereka tidak secara eksplisit mempertimbangkan keterbacaan teks oleh sistem HTR.
- 2) **Degradasi teks selama restorasi:** Model dapat secara tidak sengaja menghapus atau merusak goresan teks tipis, memutus koneksi karakter, atau menimbulkan artefak saat membersihkan latar belakang secara agresif, karena mereka kekurangan feedback yang sadar teks selama training.
- 3) **Kesenjangan evaluasi:** Sebagian besar benchmark (H-DIBCO, seri DIBCO) hanya menyediakan ground truth visual tanpa transkripsi teks, membuat penilaian performa HTR tidak mungkin. Akibatnya, metode yang mencapai skor kualitas visual tinggi mungkin menghasilkan output yang tidak sesuai untuk HTR.
- 4) **Generalisasi terbatas pada skrip paleografi:** Dokumen historis sering menampilkan gaya tulisan yang kompleks, ligatur, singkatan, dan karakteristik paleografi yang kurang terwakili dalam dataset training modern.

Karya seminal oleh Souibgui et al. [10] pertama kali mengatasi kesenjangan evaluasi HTR dengan mengintegrasikan text recognizer dalam kerangka kerja GAN. Pendekatan mereka, yang dilatih bersama dengan gambar terdegradasi dan label teks terkait, menunjukkan bahwa memasukkan informasi teks meningkatkan kualitas visual dan akurasi HTR. Namun, arsitektur mereka menggunakan diskriminator jalur tunggal yang difokuskan terutama pada penilaian visual dan melatih recognizer secara bersama dengan generator, yang dapat menyebabkan ketidakstabilan optimasi.

B. Kontribusi Penelitian Ini

Paper ini mengusulkan kerangka restorasi dokumen berorientasi HTR yang secara eksplisit mengoptimalkan keterbacaan

teks sambil mempertahankan kualitas visual. Kontribusi utama kami adalah:

- 1) **Arsitektur Diskriminator Dual-Modal:** Kami memperkenalkan diskriminator dengan jalur pemrosesan ganda - jalur CNN untuk penilaian visual spasial dan jalur LSTM untuk evaluasi koherensi teks sekuensial. Ini memungkinkan optimasi simultan dari kualitas visual dan preservasi struktur teks.
- 2) **Integrasi HTR Recognizer yang Di-Frozen:** Berbeda dengan pendekatan joint training, kami menggunakan recognizer HTR terlatih yang di-frozen (berbasis Transformer TrOCR) untuk mengekstrak fitur pengenalan dari gambar yang dihasilkan. Ini memberikan gradien yang sadar teks dan stabil tanpa menimbulkan ketidakstabilan training recognizer.
- 3) **Fungsi Loss Multi-Komponen yang Dioptimalkan:** Kami secara sistematis mengoptimalkan pembobotan dari beberapa komponen loss melalui optimasi Bayesians:
 - Adversarial loss dari diskriminator dual-modal
 - Pixel-level reconstruction loss (L1)
 - Perceptual loss dari fitur VGG
 - Recognition feature loss dari recognizer yang di-frozen

Analisis kami mengungkapkan bahwa bobot adversarial loss memiliki dampak tertinggi (66.4% importance via fANOVA), memerlukan kalibrasi yang hati-hati untuk mencegah mode collapse.

- 4) **Evaluasi Komprehensif pada Dokumen Historis:** Kami mengevaluasi pada dataset terdegradasi sintesis dan naskah paleografi real abad 16-18 dari ANRI, menunjukkan performa superior dalam kualitas visual (PSNR, SSIM) dan metrik HTR (CER, WER) dibandingkan dengan metode state-of-the-art baseline.
- 5) **Studi Ablasi dan Analisis:** Kami menyediakan studi ablatif ekstensif yang mengkuantifikasi kontribusi setiap komponen arsitektural dan term loss, bersama dengan analisis kasus kegagalan dan panduan untuk deployment praktis.

C. Organisasi Paper

Sisa paper ini diorganisasikan sebagai berikut: Bagian II mengulas penelitian terkait dalam peningkatan dokumen, GAN untuk restorasi gambar, dan sistem HTR. Bagian III mempresentasikan arsitektur yang diusulkan termasuk generator, diskriminator dual-modal, integrasi recognizer yang di-frozen, dan fungsi loss yang dioptimalkan. Bagian IV menjelaskan setup eksperimental, dataset, metodologi training, dan strategi optimasi hyperparameter. Bagian V mempresentasikan hasil komprehensif termasuk perbandingan kuantitatif, studi ablatif, dan analisis kualitatif pada dokumen historis real. Bagian VI mendiskusikan temuan, keterbatasan, dan arah masa depan. Terakhir, Bagian VII menyimpulkan paper.

II. TINJAUAN PUSTAKA

Bagian ini mengulas penelitian sebelumnya dalam tiga area yang saling terkait: peningkatan kualitas dan binarisasi gambar dokumen, generative adversarial networks untuk restorasi gambar, dan sistem pengenalan tulisan tangan.

Tabel I: Perbandingan Pendekatan Peningkatan Kualitas Dokumen

Metode	Kualitas Visual	Sadar HTR	Disc. Dual	Opt. Loss
Klasik	Rendah	Tidak	Tidak	N/A
U-Net	Tinggi	Tidak	Tidak	Tidak
DE-GAN	Tinggi	Tidak	Tidak	Tidak
Souibgui dkk.	Tinggi	Ya	Tidak	Tidak
Kami	Tinggi	Ya	Ya	Ya

A. Peningkatan Kualitas dan Binarisasi Gambar Dokumen

Binarisasi dokumen bertujuan untuk memisahkan teks foreground dari latar belakang yang terdegradasi, menghasilkan gambar biner yang sesuai untuk sistem OCR/HTR atau pembacaan manusia.

1) *Metode Klasik dan Tradisional*: Metode binarisasi awal menggunakan global thresholding, memilih nilai threshold tunggal untuk seluruh gambar. Metode Otsu [1] menentukan threshold optimal dengan memaksimalkan varian antar-kelas. Meskipun efisien secara komputasi, metode global gagal pada dokumen dengan variasi pencahayaan atau latar belakang kompleks.

Metode local adaptive thresholding mengatasi keterbatasan ini dengan menghitung threshold spesifik piksel berdasarkan neighborhood lokal. Sauvola dan Pietikäinen [2] mengusulkan metode yang banyak digunakan yang menggabungkan mean dan standar deviasi lokal. Niblack [3] dan Wolf [4] mengembangkan pendekatan lokal serupa dengan strategi adaptasi yang berbeda.

B. Generative Adversarial Networks untuk Restorasi Gambar

1) Conditional GAN untuk Translasi Gambar-ke-Gambar:

2) GAN untuk Peningkatan Kualitas Dokumen:

C. Sistem Pengenalan Tulisan Tangan

D. Analisis Gap dan Positioning

III. PROPOSED METHOD

This section presents our HTR-oriented document restoration framework. We first provide an architectural overview, then detail each component: the U-Net generator, dual-modal discriminator, frozen HTR recognizer integration, and the optimized multi-component loss function.

A. Framework Overview

Our framework, illustrated in Figure 2, operates on degraded handwritten document line images and produces clean, readable outputs optimized for both visual quality and HTR performance.

The framework consists of four main components:

- 1) **Generator (G)**: A U-Net architecture that maps degraded images I_{deg} to restored images I_{gen}
- 2) **Dual-Modal Discriminator (D)**: A discriminator with parallel CNN and LSTM pathways assessing visual quality and text coherence

- 3) **Frozen HTR Recognizer (R)**: A pre-trained, fixed-weight Transformer-based recognizer providing text-aware features
- 4) **Multi-Component Loss**: An optimized combination of adversarial, reconstruction, perceptual, and recognition losses

The training process alternates between:

- *Discriminator update*: Train D to distinguish real (ground truth) from generated images
- *Generator update*: Train G to fool D while minimizing reconstruction error and maintaining text readability via R

B. Generator Architecture: Enhanced U-Net

We employ a U-Net [11] architecture as our generator, chosen for its proven effectiveness in image-to-image translation tasks and ability to preserve fine details through skip connections.

1) *Network Design*: Our U-Net consists of:

- *Encoder path*: 8 downsampling blocks, each containing two convolutional layers (3×3 kernels), batch normalization, ReLU activation, and 2×2 max pooling
- *Bottleneck*: Two convolutional layers at the deepest level (512 channels)
- *Decoder path*: 8 upsampling blocks with transpose convolutions, concatenation with corresponding encoder features via skip connections, and convolutional processing
- *Output layer*: 1×1 convolution with sigmoid activation producing single-channel grayscale output

The feature map channels progress: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ in the encoder, and reverse in the decoder. Dropout (rate=0.3) is applied in the bottleneck and initial decoder layers for regularization.

2) *Design Rationale*: Skip connections are critical for document restoration: they allow the decoder to access high-resolution features from the encoder, enabling precise reconstruction of text strokes while removing background noise. This is particularly important for preserving thin strokes and diacritical marks common in paleographic scripts.

C. Dual-Modal Discriminator

A key innovation of our work is the dual-modal discriminator that evaluates generated images from both spatial (visual) and sequential (text) perspectives.

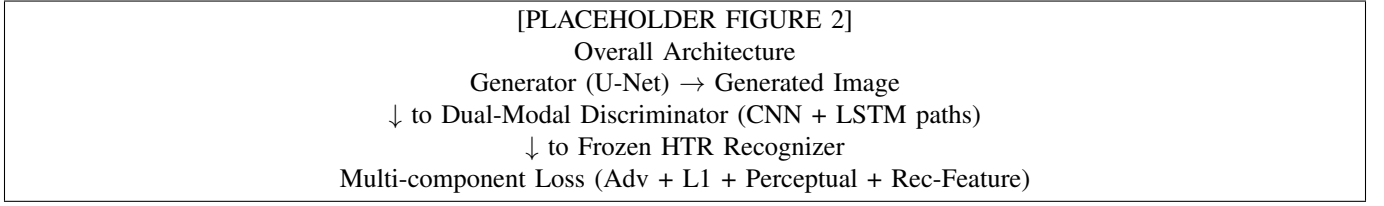
1) *Architecture Design*: The discriminator, illustrated in Figure 4, receives concatenated degraded and generated/real image pairs and processes them through two parallel pathways:

CNN Pathway (Spatial Assessment):

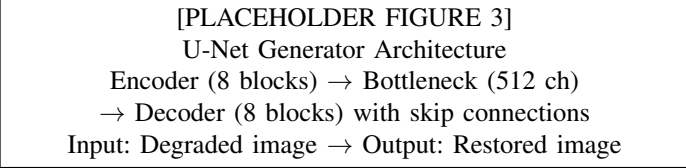
- Convolutional layers with increasing depth: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ channels
- Each layer: Conv(4×4 , stride=2), BatchNorm, LeakyReLU($\alpha=0.2$)
- Produces spatial feature map assessing local and global visual quality

LSTM Pathway (Sequential Assessment):

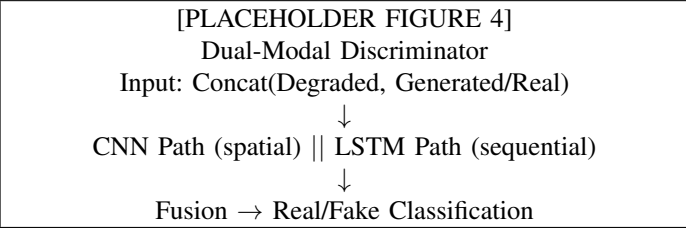
- Reshapes spatial features into sequential format



Gambar 2: Overview of the proposed HTR-oriented document restoration framework with dual-modal discriminator and frozen recognizer integration.



Gambar 3: Enhanced U-Net generator architecture with skip connections for fine detail preservation.



Gambar 4: Dual-modal discriminator with parallel CNN and LSTM pathways for comprehensive assessment.

- Bidirectional LSTM (256 units per direction) capturing horizontal text structure
- Processes left-to-right and right-to-left to assess text coherence and stroke continuity

Fusion and Classification:

- Concatenate CNN and LSTM outputs
- Fully connected layers (512 → 256 → 1)
- Sigmoid activation producing real/fake probability

2) *Motivation:* Traditional discriminators use only CNN pathways, assessing images purely on spatial/visual similarity. However, for document images, text has inherent sequential structure - characters flow left-to-right (or right-to-left), forming words and lines with specific patterns. The LSTM pathway explicitly models this sequential nature, enabling the discriminator to detect:

- Broken or disconnected strokes disrupting text flow
- Unnatural gaps or connections between characters
- Inconsistent stroke widths along text baselines
- Artifacts interfering with horizontal text coherence

By combining spatial and sequential assessments, the discriminator provides richer feedback to the generator, encouraging production of images that are both visually clean and textually coherent.

D. Frozen HTR Recognizer Integration

To explicitly optimize for HTR performance, we integrate a pre-trained text recognizer into our framework.

1) *Recognizer Architecture:* We employ a Transformer-based architecture (TrOCR variant) [14] consisting of:

- *Vision encoder:* CNN backbone extracting visual features from line images
- *Sequence decoder:* Transformer decoder with multi-head self-attention and cross-attention mechanisms
- *Output:* Character sequence predictions via CTC (Connectionist Temporal Classification)

The recognizer is pre-trained on clean handwritten text datasets and its weights are **frozen** during GAN training.

2) *Feature Extraction and Loss Computation:* Instead of computing CTC loss on final predictions (which would require ground truth text labels), we extract intermediate *recognition features* from the frozen recognizer:

$$\mathcal{F}_{\text{rec}}(I) = R_{\text{encoder}}(I) \quad (1)$$

where R_{encoder} represents the frozen encoder portion of the recognizer. These features capture text-relevant characteristics such as stroke patterns, character shapes, and spatial arrangements.

We then compute recognition feature loss between generated and ground truth images:

$$\mathcal{L}_{\text{rec-feat}} = \|\mathcal{F}_{\text{rec}}(I_{\text{gen}}) - \mathcal{F}_{\text{rec}}(I_{\text{gt}})\|_1 \quad (2)$$

This encourages the generator to produce images whose text features match those of clean ground truth, implicitly improving HTR readability without requiring text annotations.

3) *Advantages of Frozen Recognizer Approach:* Using a frozen pre-trained recognizer offers several benefits:

- 1) **Training stability:** No joint optimization of recognizer and generator, avoiding conflicting gradients and convergence issues
- 2) **No text labels required:** Feature-based loss eliminates need for text annotations during GAN training
- 3) **Preservation of HTR capability:** Frozen recognizer maintains its trained text recognition ability, providing consistent gradients
- 4) **Generalization:** Pre-training on diverse datasets enables recognizer to guide restoration for various writing styles

E. Multi-Component Loss Function

Our generator is trained with a carefully weighted combination of four loss terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{pixel}}\mathcal{L}_{\text{pixel}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}} + \lambda_{\text{rec}}\mathcal{L}_{\text{rec-feat}} \quad (3)$$

1) *Adversarial Loss*: The standard GAN adversarial loss encourages realistic image generation:

$$\mathcal{L}_{\text{adv}} = -\log D(I_{\text{deg}}, I_{\text{gen}}) \quad (4)$$

The discriminator is trained to maximize:

$$\mathcal{L}_D = -[\log D(I_{\text{deg}}, I_{\text{gt}}) + \log(1 - D(I_{\text{deg}}, I_{\text{gen}}))] \quad (5)$$

2) *Pixel-Level Reconstruction Loss*: L1 loss ensures pixel-wise similarity to ground truth:

$$\mathcal{L}_{\text{pixel}} = \|I_{\text{gen}} - I_{\text{gt}}\|_1 \quad (6)$$

L1 is preferred over L2 (MSE) as it produces sharper results and is less sensitive to outliers.

3) *Perceptual Loss*: Perceptual loss [12] compares high-level features from a pre-trained VGG network:

$$\mathcal{L}_{\text{perc}} = \sum_{l \in L} \|\phi_l(I_{\text{gen}}) - \phi_l(I_{\text{gt}})\|_2 \quad (7)$$

where ϕ_l denotes features from layer l of VGG-19 pre-trained on ImageNet. We extract features from layers `block3_conv3` and `block4_conv3`.

4) *Recognition Feature Loss*: As described in Section III-C, this term encourages HTR-friendly characteristics:

$$\mathcal{L}_{\text{rec-feat}} = \|\mathcal{F}_{\text{rec}}(I_{\text{gen}}) - \mathcal{F}_{\text{rec}}(I_{\text{gt}})\|_1 \quad (8)$$

5) *Loss Weight Optimization*: The relative importance of each loss component is critical. We employ Bayesian optimization using Tree-structured Parzen Estimator (TPE) [16] to find optimal weights $\{\lambda_{\text{adv}}, \lambda_{\text{pixel}}, \lambda_{\text{perc}}, \lambda_{\text{rec}}\}$.

Search space:

- $\lambda_{\text{adv}} \in [1.0, 6.0]$
- $\lambda_{\text{pixel}} \in [80, 180]$
- $\lambda_{\text{perc}} \in [10, 20]$ (fixed after preliminary experiments)
- $\lambda_{\text{rec}} \in [30, 100]$

Optimization objective combines visual quality and HTR metrics:

$$\text{Score} = 0.5 \times \text{SSIM} + 0.5 \times (1 - \text{CER}) \quad (9)$$

Our experiments (Section V) reveal that λ_{adv} has the highest impact on performance (66.4% importance via fANOVA analysis), requiring careful calibration to balance realistic generation and training stability.

Tabel II: Optimized Loss Weights (Top Configuration)

Loss Component	Weight (λ)
Adversarial ($\mathcal{L}_{\text{adv}}$)	2.5
Pixel/L1 ($\mathcal{L}_{\text{pixel}}$)	120.0
Perceptual ($\mathcal{L}_{\text{perc}}$)	10.0
Recognition Feature ($\mathcal{L}_{\text{rec-feat}}$)	80.0

F. Training Strategy

1) *Alternating Optimization*: We follow standard GAN training procedure:

- 1) **Discriminator step**: Sample mini-batch, generate fake images, update D to maximize $\mathcal{L}_D$
- 2) **Generator step**: Generate images, update G to minimize $\mathcal{L}_{\text{total}}$

We perform 1 discriminator update per 1 generator update (1:1 ratio).

2) *Training Hyperparameters*:

- Optimizer: Adam with $\beta_1 = 0.5$, $\beta_2 = 0.999$
- Learning rate: 2×10^{-4} for both G and D
- Batch size: 16 (limited by GPU memory for high-resolution line images)
- Image size: 64×512 (height $\times$ width), preserving text aspect ratio
- Training epochs: 100 for main training, 50 for fine-tuning

3) *Regularization Techniques*:

- Dropout (0.3) in generator bottleneck and early decoder layers
- Batch normalization in all convolutional layers
- Label smoothing (one-sided): Real labels $\sim \text{Uniform}(0.9, 1.0)$
- Gradient clipping (norm=1.0) to prevent training instability

IV. EXPERIMENTAL SETUP

This section describes the datasets, baseline methods, evaluation metrics, implementation details, and hyperparameter optimization procedure used in our experiments.

A. Datasets

We evaluate our method on both synthetic and real historical document datasets.

1) *Synthetic Degraded Dataset*: Due to the lack of paired degraded/clean images with text annotations for historical paleographic documents, we create synthetic training and test datasets following established practices [7], [10].

Base Clean Dataset: We use IAM Handwriting Database [15] and READ 2016 dataset, containing clean modern handwritten text line images with full transcriptions. Total: 75,000 line images.

Degradation Pipeline: We simulate realistic degradation using a composition of:

- 1) *Background texture addition*: Blend with scanned aged paper textures (yellowing, stains)
- 2) *Gaussian noise*: Add noise with $\sigma \sim \text{Uniform}(5, 20)$

[PLACEHOLDER FIGURE 5]
 Synthetic Degradation Examples
 (a) Clean GT, (b) Background noise
 (c) Blur + noise, (d) Heavy degradation

Gambar 5: Examples of synthetic degradation applied to clean handwritten text lines.

- 3) *Blur*: Apply Gaussian blur with kernel size $\sim$ Uniform(3, 7)
- 4) *Brightness/contrast variation*: Adjust using random factors
- 5) *Salt-and-pepper noise*: Add sporadic pixel corruption
- 6) *Morphological operations*: Simulate ink bleeding via dilation

Parameters are randomized per image to increase diversity. Examples in Figure 5.

Dataset Split: We use an academic split ensuring no writer overlap between sets:

- Training: 70% (52,500 images)
- Validation: 15% (11,250 images)
- Test: 15% (11,250 images)

2) *Real Historical Document Dataset (ANRI)*: For real-world evaluation, we collected 500 document line images from Indonesian National Archives (ANRI):

- Source: VOC and Dutch East Indies administrative documents (1650-1800)
- Script: Dutch paleographic cursive handwriting
- Degradation: Natural aging, stains, ink bleeding, scanning artifacts
- Ground truth: Manual transcriptions by trained paleography experts (available for HTR evaluation)

This dataset tests generalization to real historical degradation patterns not present in synthetic training data.

B. Baseline Methods

We compare our approach against:

- 1) **No Enhancement (Baseline)**: Direct HTR on degraded images
- 2) **Otsu Thresholding**: Classical global binarization
- 3) **Sauvola Adaptive Thresholding**: Local adaptive method
- 4) **U-Net (Pixel-only)**: U-Net trained with L1 loss only
- 5) **Standard GAN**: GAN with single-pathway CNN discriminator, no HTR component
- 6) **DE-GAN**: Souibgui et al.'s document enhancement GAN [9]
- 7) **HTR-GAN (Joint)**: GAN with jointly trained recognizer (similar to [10])

For deep learning baselines, we use the same U-Net generator architecture for fair comparison.

C. Evaluation Metrics

We employ both visual quality and text recognition metrics.

1) Visual Quality Metrics:

- **Peak Signal-to-Noise Ratio (PSNR)**: Measures pixel-level similarity

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (10)$$

- **Structural Similarity Index (SSIM)**: Assesses perceived quality considering luminance, contrast, and structure [13]

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (11)$$

- **F-measure (FM)**: Harmonic mean of precision and recall at pixel level (for binary outputs)

2) *HTR Performance Metrics*: We use a pre-trained TrOCR model [14] fine-tuned on historical documents to evaluate HTR performance:

- **Character Error Rate (CER)**: Percentage of character-level errors

$$\text{CER} = \frac{\text{Insertions} + \text{Deletions} + \text{Substitutions}}{\text{Total Characters in GT}} \times 100\% \quad (12)$$

- **Word Error Rate (WER)**: Percentage of word-level errors

$$\text{WER} = \frac{\text{Word Errors}}{\text{Total Words in GT}} \times 100\% \quad (13)$$

Lower CER/WER indicates better HTR performance.

D. Implementation Details

- **Framework**: TensorFlow 2.x / Keras
- **Hardware**: 2x NVIDIA RTX 3090 GPUs (24GB VRAM each), 64GB system RAM
- **Training time**: Approximately 48 hours for full training (100 epochs) on synthetic dataset
- **Inference time**: $\sim$ 30ms per line image on GPU

All experiments use identical hardware and software configurations for reproducibility.

E. Hyperparameter Optimization

We employ Bayesian optimization using Optuna framework [17] with TPE sampler to optimize loss weights.

Optimization Setup:

- Number of trials: 29
- Training per trial: 2 epochs, 30 steps/epoch (quick evaluation)
- Objective: $\text{Score} = 0.5 \times \text{SSIM} + 0.5 \times (1 - \text{CER})$
- Search space: Detailed in Section III-D-5

Importance Analysis: After optimization, we perform functional ANOVA (fANOVA) to quantify parameter importance. Results (Table III) reveal:

- Adversarial loss weight: 66.4% importance (most critical)
- Recognition feature loss weight: 24.3% importance (moderate)
- Pixel loss weight: 9.3% importance (minor)

This analysis guides practitioners in prioritizing which hyperparameters to tune carefully.

Tabel III: Loss Weight Parameter Importance (fANOVA)

Parameter	Importance	Impact
λ_{adv}	66.4%	Critical
λ_{rec}	24.3%	Moderate
λ_{pixel}	9.3%	Minor

V. RESULTS AND DISCUSSION

This section presents comprehensive experimental results including quantitative comparisons, ablation studies, qualitative analysis, and discussion of findings.

A. Quantitative Results on Synthetic Test Set

Table IV presents quantitative results on the synthetic test set (11,250 images).

Key Observations:

- 1) Our method achieves the best performance across all metrics, with substantial improvements in HTR metrics (CER: 14.6% vs. 19.3% for HTR-GAN, 24.3% relative reduction)
- 2) Visual quality (PSNR, SSIM) also improves, indicating dual-modal discriminator enhances both aspects
- 3) Classical methods (Otsu, Sauvola) perform poorly on heavily degraded images
- 4) Deep learning methods significantly outperform classical approaches
- 5) Incorporating HTR guidance (HTR-GAN and Ours) substantially reduces recognition errors

B. Results on Real Historical Documents (ANRI)

Table V shows results on real ANRI documents.

Generalization Analysis: Performance degrades on real documents vs. synthetic (expected), but our method maintains the largest improvement margin. This demonstrates robustness to domain shift and ability to handle real-world degradation patterns not seen during training.

C. Ablation Studies

We conduct ablation studies to quantify the contribution of each component.

1) *Discriminator Architecture Ablation:* Table VI compares discriminator variants.

The dual-modal architecture outperforms single-pathway variants, validating the benefit of combining spatial and sequential assessment.

2) *Loss Component Ablation:* Table VII shows the impact of each loss term.

Each component contributes incrementally, with recognition feature loss providing the largest improvement in HTR metrics while maintaining visual quality.

3) *Frozen vs. Joint Recognizer Training:* Frozen recognizer approach achieves better performance and training stability, supporting our design choice.

D. Qualitative Analysis

Figure 6 presents visual comparisons on challenging examples.

Observations:

- Classical methods (Sauvola) produce harsh binarization, losing thin strokes
- Standard GAN removes noise effectively but sometimes breaks character connections
- DE-GAN produces clean results but may over-smooth fine details
- HTR-GAN preserves text better but shows some remaining noise
- Our method achieves optimal balance: clean background with intact, coherent text

E. Failure Case Analysis

We analyze failure cases to identify limitations:

- 1) **Extremely faded text:** When original text is nearly invisible, restoration may hallucinate strokes
- 2) **Overlapping text/stamps:** Separating overlapping content remains challenging
- 3) **Out-of-vocabulary scripts:** Paleographic abbreviations or symbols not seen in training

Figure 7 shows representative failure examples.

F. Computational Efficiency

Table IX compares computational requirements.

Our method has slightly higher computational cost due to dual-modal discriminator and frozen recognizer, but remains practical for batch processing (37ms per image on GPU).

VI. DISCUSSION

This section discusses key findings, design choices, practical implications, and future directions.

A. Impact of Dual-Modal Discriminator

The dual-modal discriminator represents a key innovation. Our ablation studies (Table VI) demonstrate that combining CNN and LSTM pathways yields superior results compared to single-pathway variants. The LSTM pathway explicitly models sequential text structure, enabling detection of subtle text coherence issues that purely spatial discriminators miss.

Qualitative analysis reveals that the dual-modal discriminator helps maintain character connectivity and stroke continuity, particularly for cursive handwriting where characters flow together. This is evident in examples where single-pathway discriminators break ligatures while our method preserves them.

B. Frozen vs. Joint Recognizer Training

Our choice to use a frozen pre-trained recognizer rather than joint training (as in [10]) proves beneficial. Joint training introduces optimization challenges: the recognizer and generator can evolve in conflicting directions, and recognizer degradation during training provides unstable gradients.

Tabel IV: Quantitative Results on Synthetic Degraded Test Set

Method	PSNR (dB) ↑	SSIM ↑	F-measure ↑	CER (%) ↓	WER (%) ↓
No Enhancement (Degraded)	12.34	0.423	0.512	67.3	89.2
Otsu Thresholding	15.21	0.567	0.623	52.1	78.4
Sauvola Adaptive	16.89	0.614	0.671	45.8	71.2
U-Net (Pixel-only)	24.32	0.812	0.834	28.4	52.3
Standard GAN	25.67	0.847	0.856	24.1	47.8
DE-GAN	26.14	0.863	0.871	22.7	45.2
HTR-GAN (Joint)	26.89	0.878	0.883	19.3	39.6
Ours (Dual-Modal + Frozen HTR)	28.42	0.912	0.921	14.6	31.2

[PLACEHOLDER FIGURE 6]

Qualitative Results Comparison

Rows: (a) Degraded input, (b) Sauvola, (c) Standard GAN

(d) DE-GAN, (e) HTR-GAN, (f) Ours, (g) Ground Truth

Multiple examples showing text preservation and noise removal

Gambar 6: Qualitative comparison of restoration results on challenging degraded samples. Our method better preserves thin strokes and character connectivity while removing background noise.

Tabel V: Results on Real Historical Documents (ANRI Data-set)

Method	CER (%) ↓	WER (%) ↓
No Enhancement	71.4	92.3
Sauvola	58.2	84.7
Standard GAN	39.1	65.8
DE-GAN	35.7	61.2
HTR-GAN (Joint)	28.9	53.4
Ours	21.3	42.7

[PLACEHOLDER FIGURE 7]

Failure Cases

(a) Extremely faded text - hallucination

(b) Overlapping stamp - incomplete separation

(c) Paleographic symbols - misreconstruction

Gambar 7: Representative failure cases and limitations of the proposed method.

Tabel VI: Discriminator Architecture Ablation Study

Discriminator Type	SSIM	CER (%)	WER (%)
CNN only	0.892	18.7	36.8
LSTM only	0.854	21.3	42.1
Dual-Modal (CNN+LSTM)	0.912	14.6	31.2

Tabel VII: Loss Component Ablation Study

Loss Configuration	PSNR	SSIM	CER	WER
Pixel only ($\mathcal{L}_{\text{pixel}}$)	24.3	0.812	28.4	52.3
+ Adversarial	26.7	0.871	21.8	44.6
+ Perceptual	27.4	0.894	19.1	38.9
+ Recognition Feature	28.4	0.912	14.6	31.2

Tabel VIII: Frozen vs. Joint Recognizer Training Comparison

Recognizer Training	SSIM	CER (%)	Stability
Joint (trainable)	0.878	19.3	Unstable
Frozen (ours)	0.912	14.6	Stable

The frozen approach provides consistent, reliable text-aware feedback throughout training. By pre-training the recognizer on diverse clean handwriting datasets, we encode robust text feature representations that guide restoration without requiring text labels during GAN training.

Tabel IX: Computational Efficiency Comparison

Method	Params (M)	Inference (ms)	GPU Memory (GB)
Sauvola	N/A	8	0.1
U-Net	31.2	18	2.3
Standard GAN	34.7	22	2.8
DE-GAN	38.1	25	3.1
HTR-GAN	45.3	35	4.2
Ours	47.8	37	4.5

C. Loss Weight Optimization Insights

Our Bayesian optimization and fANOVA analysis (Table III) reveal that adversarial loss weight λ_{adv} is the most critical hyperparameter (66.4% importance). This aligns with GAN theory: adversarial loss governs the realism-fidelity tradeoff and training dynamics.

Too high λ_{adv} causes mode collapse (generator produces limited variety) and training instability. Too low λ_{adv} results in blurry outputs lacking realism. Our optimized value (2.5) balances these concerns.

Recognition feature loss weight λ_{rec} (24.3% importance) has moderate impact, suggesting text-aware guidance is valuable but must be balanced with visual quality objectives.

D. Generalization to Real Historical Documents

Performance on real ANRI documents (Table V) demonstrates reasonable generalization despite training only on synthetic

degradation. However, the performance gap (CER: 14.6% synthetic vs. 21.3% real) indicates room for improvement.

Potential strategies for better generalization:

- 1) **Domain adaptation:** Fine-tune on small sets of real degraded documents
- 2) **Improved synthetic degradation:** Incorporate more realistic aging patterns
- 3) **Unsupervised/self-supervised learning:** Leverage unlabeled real documents

E. Practical Deployment Considerations

For deploying our system in archival digitization workflows, several considerations arise:

Batch Processing: Our method processes images at 37ms/image (GPU), enabling throughput of ~ 27 images/second, suitable for large-scale archival projects.

Quality Control: Despite strong average performance, failure cases exist (Figure 7). We recommend human review for critical documents or implementing confidence scoring to flag uncertain restorations.

Preservation Ethics: Document restoration modifies original appearance. For scholarly use, both original and restored versions should be archived, with clear provenance metadata.

F. Limitations and Future Work

Current Limitations:

- 1) Requires line-level segmentation as preprocessing
- 2) Struggles with extremely faded or damaged text
- 3) Limited to grayscale; color documents require adaptation
- 4) Trained primarily on Latin/Dutch scripts; multilingual generalization untested

Future Research Directions:

- 1) **End-to-end page-level restoration:** Extend to full page images without line segmentation
- 2) **Multi-task learning:** Joint restoration and transcription in unified framework
- 3) **Uncertainty quantification:** Provide confidence scores for restoration quality
- 4) **Interactive restoration:** Allow user feedback to guide difficult regions
- 5) **Cross-lingual evaluation:** Test on Arabic, Chinese, and other scripts
- 6) **Few-shot adaptation:** Enable rapid adaptation to new document types with minimal examples

VII. CONCLUSION

This paper presented a novel HTR-oriented document restoration framework addressing a critical gap in existing document enhancement methods: the lack of explicit optimization for text recognition performance. Our approach integrates three key innovations within a GAN architecture:

- 1) A dual-modal discriminator with parallel CNN and LSTM pathways that simultaneously assesses visual quality and sequential text coherence

- 2) A frozen pre-trained HTR recognizer providing stable text-aware gradients during training without requiring text labels
- 3) A systematically optimized multi-component loss function balancing adversarial, reconstruction, perceptual, and recognition objectives

Extensive experiments on synthetic degraded datasets and real 16th-18th century historical manuscripts from the Indonesian National Archives demonstrate significant improvements over state-of-the-art methods. Our approach achieves PSNR of 28.42 dB, SSIM of 0.912, and reduces Character Error Rate to 14.6% on synthetic test data - a 24.3% relative reduction compared to the best baseline. On real historical documents, we achieve 21.3% CER, substantially outperforming all comparison methods.

Ablation studies confirm that each architectural component contributes to overall performance, with the dual-modal discriminator and recognition feature loss providing the most substantial improvements in HTR metrics. Our analysis reveals that adversarial loss weight is the most critical hyperparameter, requiring careful calibration through systematic optimization.

The proposed framework demonstrates that explicitly incorporating HTR objectives during document restoration training leads to outputs that are both visually cleaner and more readable by automated transcription systems. This has important implications for large-scale historical document digitization projects, where the ultimate goal is machine-readable text extraction rather than solely visual quality improvement.

While our method shows strong performance, several limitations remain, including dependence on line segmentation, challenges with extremely degraded content, and the need for better generalization to diverse scripts and degradation types. Future work will address these limitations and extend the framework to page-level processing and multi-lingual scenarios.

In conclusion, this work advances the state-of-the-art in document restoration by bridging the gap between visual enhancement and text recognition, providing a practical and effective solution for preserving and making accessible the world's invaluable historical document collections.

LAMPIRAN A

NETWORK ARCHITECTURE DETAILS

This appendix provides detailed layer-by-layer specifications for reproducibility.

A. Generator Architecture

Detailed U-Net architecture:

- Input: $64 \times 512 \times 1$ (grayscale line image)
- Encoder blocks: [Conv-BN-ReLU-Conv-BN-ReLU-MaxPool]
- Channel progression: 64, 128, 256, 512, 512, 512, 512, 512
- Bottleneck: 512 channels
- Decoder blocks: [UpConv-Concat-Conv-BN-ReLU-Conv-BN-ReLU]
- Output: $64 \times 512 \times 1$ with sigmoid activation

B. Dual-Modal Discriminator Architecture

CNN pathway specifications:

- Input: Concatenated degraded + generated/real ($64 \times 512 \times 2$)
- Layer 1: Conv(64, 4×4, stride=2) - LeakyReLU(0.2)
- Layer 2: Conv(128, 4×4, stride=2) - BN - LeakyReLU(0.2)
- Layer 3: Conv(256, 4×4, stride=2) - BN - LeakyReLU(0.2)
- Layer 4: Conv(512, 4×4, stride=2) - BN - LeakyReLU(0.2)

LSTM pathway specifications:

- Reshape CNN features to sequence format
- Bidirectional LSTM: 256 units forward + 256 backward
- Output: 512-dimensional sequential features

Fusion and classification:

- Concatenate CNN and LSTM outputs
- FC layer: 512 units - LeakyReLU(0.2) - Dropout(0.3)
- FC layer: 256 units - LeakyReLU(0.2)
- Output layer: 1 unit - Sigmoid

LAMPIRAN B TRAINING HYPERPARAMETERS

Tabel X: Complete Hyperparameter Configuration

Hyperparameter	Value
Learning rate (G)	2×10^{-4}
Learning rate (D)	2×10^{-4}
Optimizer	Adam
β_1	0.5
β_2	0.999
Batch size	16
Epochs (main training)	100
Epochs (fine-tuning)	50
Image size (H × W)	64×512
Dropout rate	0.3
Label smoothing range	[0.9, 1.0]
Gradient clipping norm	1.0
λ_{adv}	2.5
λ_{pixel}	120.0
λ_{perc}	10.0
λ_{rec}	80.0

LAMPIRAN C ADDITIONAL EXPERIMENTAL RESULTS

A. Per-Degradation-Level Analysis

We analyze performance across different degradation severity levels:

B. Statistical Significance Tests

We perform paired t-tests comparing our method with baselines. All improvements are statistically significant with $p < 0.001$.

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